

[illegible]

Pre-mortem analysis record

Severity: This column lists the severity rating of each of the failures listed in column #2. Use the failure effects listed in column #3 to determine the rating. The typical scale of severity is from zero to ten; zero being the least severe, while ten is the most severe consequence(s).

Root cause: For each failure listed in column #2, root cause analysis is done to find an answer to the question—what will cause this step to go wrong?

Occurrence: This column is another rating based on the frequency of failure. How frequently are these failures, as listed in column #2, likely to occur? Occurrence is ranked on a scale of one to ten, where one is a low occurrence, and ten is a high or frequent occurrence.

Controls: An answer to the question—What controls are in place currently to prevent potential failure as per column #2? What controls are in place to detect the occurrence of a fault, if any?

Detection: This is another rating column where ease of detection of each failure is assessed. Typical questions to ask are: How easy is it to detect each of the potential failures? What is the likelihood that you can discover these potential failures promptly or before they reach the customers? Detection is ranked on a scale of ten to one (please note reversal of the scale). Here rating of one means easily and quickly detectable failure, whereas ten means unlikely and very late detection of failure. Late detection often means a more problematic situation, and therefore the rating for late detection is higher.

RPN (Risk Priority Number):
The risk priority is determined by multiplying all the three ratings from column #4, 6, and 8. So, $RPN = \text{severity} \times \text{occurrence} \times \text{detection}$. Thus, a high RPN would indicate a high-risk process step or solution function (as in column #1). Accordingly, steps or functions with higher RPN warrant immediate attention for fixing.

Recommended actions: In this column, SMEs would recommend one or more actions to handle the risks identified. These actions may be directed towards reducing the severity, chances of failure occurrence, improving the detection level, or maybe all of the above.

Action owner and target date: This column is essential from the project management point of view as well as for tracking. Each recommended action can be assigned to a specific owner and carried out before the target date to contain the risks.

Actions completed: This column lists all the actions taken, recommended or otherwise, to lower the risk level (RPN) to an acceptable level or lower.

New severity: Once the actions listed in column #12 are complete, the same exercise must be repeated to arrive at a new level of severity.

New occurrence: The occurrence must have changed depending on actions completed, so this column has a new occurrence rating.